

NETWORKED CONTROL

03/07/2019

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EXERCIZES

1) Explain how, qualitatively, the choice of the sampling time affects the performance of a networked control system, in case the control network is affected by delays and possible packet dropouts. *See the slides.*

2) Consider the following continuous-time system $\mathcal{S}$.

$$\begin{cases} \dot{x}(t) &= (\log_e 2)x(t) + 3 (\log_e 2)u(t) \\ y(t) &= 3x(t) \end{cases}$$

- A) Assume that the system is controlled using a continuous-time control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.
- B) Compute the parameters F, G , and H of the corresponding discrete-time dynamical equation

$$x_{k+1} = Fx_k + Gu_k, \quad y_k = Hx_k$$

with sampling time $h = 1$ s.

Also, assuming that the system is controlled using a digital control law of the type $u_k = Ky_k$, define the range of values of K that make the closed-loop system asymptotically stable.

- C) Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

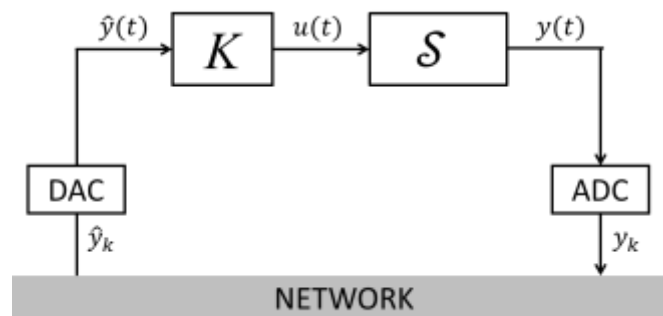


Figure 1: Networked control system

Answer the following questions considering the case where $h = 1$ s and $K = -\frac{1}{6}$.

- Is the NCS asymptotically stable in case the delay is $\tau = 0$ s and in absence of packet dropouts?
- Is the NCS asymptotically stable in case the delay is $\tau = 0.2$ s and in absence of packet dropouts?
- Assume now that the NCS is subject to packet dropouts but the delay is absent, i.e., $\tau = 0$ s. Variable θ_k has the following periodic behavior.

k	1	2	3	4	5	6	7	8	9	...
θ_k	0	0	1	0	0	1	0	0	1	...

Is the corresponding NCS asymptotically stable?

SOLUTIONS

A) Setting $u(t) = Ky(t) = 3Kx(t)$, the closed loop dynamics is described by the equation

$$\dot{x} = (\log_e 2) (1 + 9K)x$$

whose eigenvalue is $\lambda = (\log_e 2) (1 + 9K)$. The system is closed-loop asymptotically stable for $K < -1/9$.

B) The corresponding discrete-time system is

$$\begin{cases} x_{k+1} &= 2x_k + 3u_k \\ y_k &= 3x_k \end{cases}$$

Setting $u_k = Ky_k = 3Kx_k$, the closed loop dynamics is described by the equation

$$x_{k+1} = (2 + 9K)x_k$$

whose eigenvalue is $\lambda = 2 + 9K$. The system is closed-loop asymptotically stable iff $|2 + 9K| < 1$, i.e., iff $-1/3 < K < -1/9$.

C) We set $K = -1/6$ and $h = 1$ s.

- The NCS is asymptotically stable in case delays and packet dropouts are absent, in view of the response to answer B. since $-1/3 < K = -1/6 < -1/9$.
- In case packet dropouts are absent, but a delay is present, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h, \tau) = \begin{bmatrix} e^{(\log_e 2)h} + \frac{\Gamma(h - \tau)9K \log_e 2}{3} & e^{(\log_e 2)(h - \tau)} \frac{\Gamma(\tau)3K \log_e 2}{0} \\ 0 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{(\log_e 2)v} dv = \frac{1}{\log_e 2} (2^\tau - 1)$. In case $h = 1$ s, $\tau = 0.2$ s, and $K = -1/6$,

$$\Phi = \begin{bmatrix} 2 + 9K(2^{0.8} - 1) & 3K(2 - 2^{0.8}) \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 0.888 & -0.129 \\ 3 & 0 \end{bmatrix}$$

whose eigenvalues are $\lambda_1 = 0.4441 + 0.4356j$ and $\lambda_2 = 0.4441 - 0.4356j$, whose modulus is equal to $0.6221 < 1$. The NCS is therefore asymptotically stable.

- To analyze the stability of the NCS under packet dropouts (but in absence of delays) defined by the sequence of values of variable θ_k , we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\tilde{\Phi}_1 = \Phi(h, \tau) = \Phi(1, 0) = \begin{bmatrix} e^{(\log_e 2)^1} + \frac{\Gamma(1)9K \log_e 2}{3} & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 3 & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{(\log_e 2)^1} & \Gamma(1)3K \log_e 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\tilde{\Phi}_1$ has eigenvalues in $\lambda_1 = 0.5$ and $\lambda_2 = 0$, while $\tilde{\Phi}_0$ has eigenvalues in 2 and 1. Since the sequence of values of θ_k is 1,0,0,1,0,0,1,0,0:

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{3k+3} = \tilde{\Phi}_1 \tilde{\Phi}_0 \tilde{\Phi}_0 z_{3k}$$

$$z_{3k+4} = \tilde{\Phi}_0 \tilde{\Phi}_0 \tilde{\Phi}_1 z_{3k+1}$$

$$z_{3k+5} = \tilde{\Phi}_0 \tilde{\Phi}_1 \tilde{\Phi}_0 z_{3k+2}$$

Therefore, $\tilde{\Phi}_1 \tilde{\Phi}_0 \tilde{\Phi}_0$, $\tilde{\Phi}_0 \tilde{\Phi}_0 \tilde{\Phi}_1$ and $\tilde{\Phi}_0 \tilde{\Phi}_1 \tilde{\Phi}_0$ can be regarded as the three-steps transition matrices for the NCS state. We resort to the fact that $\tilde{\Phi}_1 \tilde{\Phi}_0 \tilde{\Phi}_0$, $\tilde{\Phi}_0 \tilde{\Phi}_0 \tilde{\Phi}_1$ and $\tilde{\Phi}_0 \tilde{\Phi}_1 \tilde{\Phi}_0$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\tilde{\Phi}_1 \tilde{\Phi}_0 \tilde{\Phi}_0 = \begin{bmatrix} 2 & -3/4 \\ 12 & -9/2 \end{bmatrix}$$

whose eigenvalues are $-5/2$ and 0 . The NCS is therefore unstable.

3) Consider a networked control system with an ideal communication channel (i.e., no delays or packet dropouts) with bandwidth B and with signal-to-noise ratio S/N .

A) Define briefly the *channel capacity* $R_{\max}$. How do you compute $R_{\max}$?

B) In case a digital quantized controller is used, where the scalar input variable u_k is transmitted over a communication channel with rate R and in packets of length N_B bits, and where the system eigenvalues are $\lambda_1, \dots, \lambda_n$, what is the main tradeoff to satisfy in the choice of N_B and of the sampling time h ? Motivate adequately the response.

C) In case a digital quantized controller is used to control the system in Exercise 2, what is the minimum number of bits N_B encoding the control input required to make the system boundable?

SOLUTION

For points A and B see the lecture notes. For point C, the inequality is

$$\frac{N_B}{h} = N_B \geq \log_e 2 \cdot \log_2 e = 1$$